Appendix A

R Refresher

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> A concise refresher on R fundamentals for readers who need a quick review.

This appendix provides a quick reference for the R features used most often in this book. It is not a substitute for a full R tutorial but should be enough to remind you of the essentials. For a deeper treatment, see Wickham et al. (2023).

A.1 Data structures

A.1.1 Vectors

Vectors are the fundamental data type in R. Every element must share the same type (logical, integer, double, character).

```
# Numeric vector
payoffs <- c(3, 0, 5, 1)

# Character vector
strategies <- c("Cooperate", "Defect")

# Logical vector
is_dominated <- c(FALSE, TRUE, FALSE)

# Sequence shortcuts
```

```
indices <- 1:10
grid <- seq(0, 1, by = 0.1)

# Vectorised arithmetic
payoffs * 2
```

```
#> [1]  6  0 10  2
```

```
payoffs > 2
```

```
#> [1]  TRUE FALSE  TRUE FALSE
```

A.1.2 Matrices

Matrices are two-dimensional vectors. They are central to payoff representations throughout this book.

```
# Create a 2x2 payoff matrix (filled by column by default)
A <- matrix(c(3, 5, 0, 1), nrow = 2, byrow = TRUE)
rownames(A) <- c("Cooperate", "Defect")
colnames(A) <- c("Cooperate", "Defect")
A
```

```
#>           Cooperate Defect
#> Cooperate         3      5
#> Defect            0      1
```

```
# Matrix indexing
A[1, 2]       # row 1, column 2
```

```
#> [1] 5
```

```
A["Defect", ] # entire row by name
```

```
#> Cooperate    Defect
#>         0         1
```

```
# Dimensions
dim(A)
```

```
#> [1] 2 2
```

```
nrow(A)
```

```
#> [1] 2
```

```
ncol(A)
```

```
#> [1] 2
```

A.1.3 Lists

Lists can hold elements of different types and different lengths. They are used throughout the book to bundle game components together.

```
game <- list(
  players  = c("Player 1", "Player 2"),
  payoff_1 = matrix(c(3, 5, 0, 1), nrow = 2, byrow = TRUE),
  payoff_2 = matrix(c(3, 0, 5, 1), nrow = 2, byrow = TRUE)
)

# Access by name
game$players
```

```
#> [1] "Player 1" "Player 2"
```

```
game[["payoff_1"]]
```

```
#>      [,1] [,2]
#> [1,]    3    5
#> [2,]    0    1
```

```
# Access by position
game[[1]]
```

```
#> [1] "Player 1" "Player 2"
```

A.1.4 Data frames and tibbles

Data frames (and their tidyverse counterpart, tibbles) are the primary structure for tabular data.

```
# Base R data frame
df <- data.frame(
  strategy = c("Cooperate", "Defect"),
  payoff   = c(3, 1)
)

# Tidyverse tibble
tbl <- tibble(
  strategy = c("Cooperate", "Defect"),
  payoff   = c(3, 1)
)

# Tibbles print more informatively
tbl
```

```
#> # A tibble: 2 x 2
#>   strategy  payoff
#>   <chr>      <dbl>
#> 1 Cooperate      3
#> 2 Defect         1
```

A.2 Control flow

A.2.1 Conditionals

```
x <- 5

if (x > 3) {
  result <- "high"
} else if (x > 1) {
  result <- "medium"
} else {
  result <- "low"
}
result
```

```
#> [1] "high"
```

```r
# Vectorised conditional
ifelse(c(3, 0, 5, 1) > 2, "above", "below")
```

```
#> [1] "above" "below" "above" "below"
```

A.2.2 Loops

```r
# for loop
total <- 0
for (i in 1:5) {
  total <- total + i
}
total
```

```
#> [1] 15
```

```r
# while loop
count <- 0
value <- 1
while (value < 100) {
  value <- value * 2
  count <- count + 1
}
count
```

```
#> [1] 7
```

A.2.3 The apply family

The `apply` functions replace explicit loops with concise functional calls.

```r
A <- matrix(1:12, nrow = 3)

# Apply across rows (margin = 1)
apply(A, 1, sum)
```

```
#> [1] 22 26 30
```

```r
# Apply across columns (margin = 2)
apply(A, 2, mean)
```

```
#> [1]  2  5  8 11
```

```r
# sapply returns a vector; lapply returns a list
sapply(1:5, function(x) x^2)
```

```
#> [1]  1  4  9 16 25
```

```r
lapply(1:3, function(x) rep(x, x))
```

```
#> [[1]]
#> [1] 1
#>
#> [[2]]
#> [1] 2 2
#>
#> [[3]]
#> [1] 3 3 3
```

```r
# vapply is like sapply but with a specified return type
vapply(1:5, function(x) x^2, numeric(1))
```

```
#> [1]  1  4  9 16 25
```

A.3 Functions

A.3.1 Defining functions

```r
# A function that computes expected payoff
expected_payoff <- function(payoff_vec, prob_vec) {
  sum(payoff_vec * prob_vec)
}

expected_payoff(c(3, 0), c(0.6, 0.4))
```

```
#> [1] 1.8
```

```r
# Default arguments
greet <- function(name, greeting = "Hello") {
  paste(greeting, name)
}
greet("Alice")
```

```
#> [1] "Hello Alice"
```

```r
greet("Bob", greeting = "Hi")
```

```
#> [1] "Hi Bob"
```

A.3.2 Closures and environments

A closure is a function that captures variables from its enclosing environment. This pattern is useful for creating parameterised game constructors.

```r
make_discounter <- function(delta) {
  # delta is captured in the closure
  function(payoffs) {
    n <- length(payoffs)
    weights <- delta^(seq_len(n) - 1)
    sum(payoffs * weights)
  }
}

discount_95 <- make_discounter(0.95)
discount_95(c(3, 3, 3, 3, 3))
```

```
#> [1] 13.6
```

```r
# The enclosed value of delta persists
discount_50 <- make_discounter(0.50)
discount_50(c(3, 3, 3, 3, 3))
```

```
#> [1] 5.81
```

A.3.3 Anonymous functions

R 4.1+ supports the shorthand `\(x)` syntax for anonymous (lambda) functions.

```
sapply(1:5, \(x) x^2)
```

```
#> [1]  1  4  9 16 25
```

```
# Equivalent to the older syntax:
sapply(1:5, function(x) x^2)
```

```
#> [1]  1  4  9 16 25
```

A.4 Tidyverse basics

The tidyverse is a collection of packages for data wrangling and visualisation. This book uses it extensively.

A.4.1 The pipe operator

The pipe `|>` (base R 4.1+) or `%>%` (magrittr) passes the left-hand side as the first argument to the right-hand side.

```
c(4, 1, 7, 2, 9) |>
  sort() |>
  rev() |>
  head(3)
```

```
#> [1] 9 7 4
```

A.4.2 dplyr verbs

The five core verbs handle the vast majority of data manipulation tasks.

```
game_results <- tibble(
  round    = 1:6,
  player_1 = c("C", "D", "C", "C", "D", "D"),
  player_2 = c("C", "C", "D", "C", "D", "C"),
  payoff_1 = c(3, 5, 0, 3, 1, 5),
  payoff_2 = c(3, 0, 5, 3, 1, 0)
)

# filter: keep rows matching a condition
game_results |> filter(player_1 == "C")
```

```
#> # A tibble: 3 x 5
#>   round player_1 player_2 payoff_1 payoff_2
#>   <int> <chr>    <chr>       <dbl>    <dbl>
#> 1     1 C        C               3        3
#> 2     3 C        D               0        5
#> 3     4 C        C               3        3
```

```
# mutate: create or modify columns
game_results |> mutate(total = payoff_1 + payoff_2)
```

```
#> # A tibble: 6 x 6
#>   round player_1 player_2 payoff_1 payoff_2 total
#>   <int> <chr>    <chr>       <dbl>    <dbl> <dbl>
#> 1     1 C        C               3        3     6
#> 2     2 D        C               5        0     5
#> 3     3 C        D               0        5     5
#> 4     4 C        C               3        3     6
#> 5     5 D        D               1        1     2
#> 6     6 D        C               5        0     5
```

```
# select: choose columns
game_results |> select(round, payoff_1, payoff_2)
```

```
#> # A tibble: 6 x 3
#>   round payoff_1 payoff_2
#>   <int>    <dbl>    <dbl>
#> 1     1        3        3
#> 2     2        5        0
#> 3     3        0        5
#> 4     4        3        3
#> 5     5        1        1
#> 6     6        5        0
```

```
# arrange: reorder rows
game_results |> arrange(desc(payoff_1))
```

```
#> # A tibble: 6 x 5
#>   round player_1 player_2 payoff_1 payoff_2
#>   <int> <chr>    <chr>       <dbl>    <dbl>
#> 1     2 D        C               5        0
#> 2     6 D        C               5        0
#> 3     1 C        C               3        3
#> 4     4 C        C               3        3
```

```
#> 5     5 D        D                1        1
#> 6     3 C        D                0        5
```

```
# summarise (often with group_by)
game_results |>
  group_by(player_1) |>
  summarise(
    mean_payoff = mean(payoff_1),
    n_rounds    = n()
  )
```

```
#> # A tibble: 2 x 3
#>   player_1 mean_payoff n_rounds
#>   <chr>          <dbl>    <int>
#> 1 C               2           3
#> 2 D               3.67        3
```

A.4.3 Pivoting

Converting between wide and long formats is essential for plotting.

```
# Wide to long
game_long <- game_results |>
  pivot_longer(
    cols      = starts_with("payoff"),
    names_to  = "player",
    values_to = "payoff"
  )
head(game_long)
```

```
#> # A tibble: 6 x 5
#>   round player_1 player_2 player   payoff
#>   <int> <chr>    <chr>    <chr>     <dbl>
#> 1     1 C        C        payoff_1      3
#> 2     1 C        C        payoff_2      3
#> 3     2 D        C        payoff_1      5
#> 4     2 D        C        payoff_2      0
#> 5     3 C        D        payoff_1      0
#> 6     3 C        D        payoff_2      5
```

```
# Long to wide
game_long |>
  pivot_wider(names_from = player, values_from = payoff)
```

```
#> # A tibble: 6 x 5
#>   round player_1 player_2 payoff_1 payoff_2
#>   <int> <chr>    <chr>       <dbl>    <dbl>
#> 1     1 C        C               3        3
#> 2     2 D        C               5        0
#> 3     3 C        D               0        5
#> 4     4 C        C               3        3
#> 5     5 D        D               1        1
#> 6     6 D        C               5        0
```

A.4.4 ggplot2 layers

The grammar of graphics builds plots layer by layer: data, aesthetics, geometries, and scales.

```
game_results |>
  pivot_longer(
    cols = starts_with("payoff"),
    names_to = "player",
    values_to = "payoff"
  ) |>
  ggplot(aes(x = round, y = payoff, colour = player)) +
  geom_line(linewidth = 0.8) +
  geom_point(size = 2) +
  scale_colour_manual(
    values = c(payoff_1 = "#1b9e77", payoff_2 = "#d95f02"),
    labels = c("Player 1", "Player 2")
  ) +
  labs(x = "Round", y = "Payoff", colour = NULL) +
  theme_minimal()
```

A.5 Useful idioms for this book

A handful of patterns recur throughout the chapters.

```
# Replicate a simulation many times
results <- replicate(1000, {
  sample(c("Heads", "Tails"), 1)
})
table(results)
```

```
#> results
```

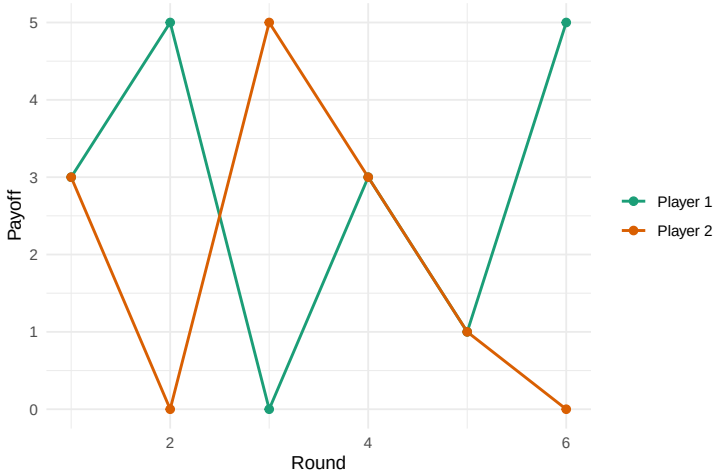


Figure A.1: Payoffs across six rounds of a repeated game.

```
#> Heads Tails
#>   499   501
```

```
# Outer product for payoff grids
p <- seq(0, 1, by = 0.25)
q <- seq(0, 1, by = 0.25)
payoff_grid <- outer(p, q, function(pi, qi) 3 * pi * qi + 1 * (1
 ↪  - pi))
payoff_grid
```

```
#>      [,1]  [,2] [,3] [,4] [,5]
#> [1,] 1.00 1.000 1.00 1.00  1.0
#> [2,] 0.75 0.938 1.12 1.31  1.5
#> [3,] 0.50 0.875 1.25 1.62  2.0
#> [4,] 0.25 0.812 1.38 1.94  2.5
#> [5,] 0.00 0.750 1.50 2.25  3.0
```

```
# Named vectors for readable code
strategy_names <- c(C = "Cooperate", D = "Defect")
strategy_names["C"]
```

```
#>           C
#> "Cooperate"
```

```r
# Setting seeds for reproducibility
set.seed(42)
sample(1:100, 5)
```

```
#> [1] 49 65 25 74 18
```